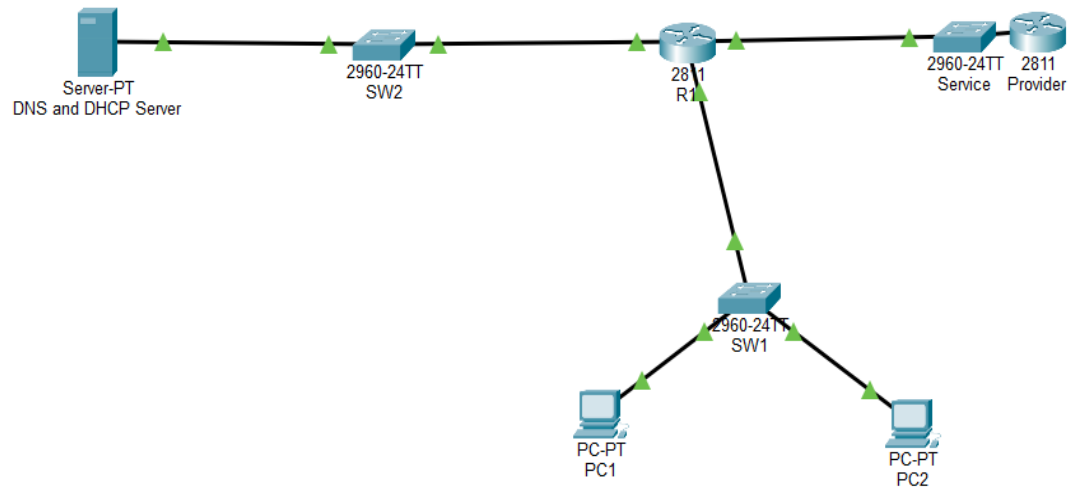


Week 10: DHCP Configuration (5 Marks)



97.

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#Interface FastEthernet0/0
Router(config-if)#ip address 10.10.30.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#ip dhcp excluded-address 10.10.30.1
Router(config)#ip dhcp pool R1-OUTSIDE
Router(dhcp-config)#network 10.10.30.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.30.1
Router(dhcp-config)#exit
Router(config)#
```

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address dhcp
Router(config-if)#no
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address 10.10.30.2, mask
255.255.255.0, hostname R1

shutdown
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.10.30.2      YES DHCP    up          up
FastEthernet0/1 10.10.10.1      YES manual  up          up
FastEthernet1/0 10.10.20.1      YES manual  up          up
FastEthernet1/1 unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down

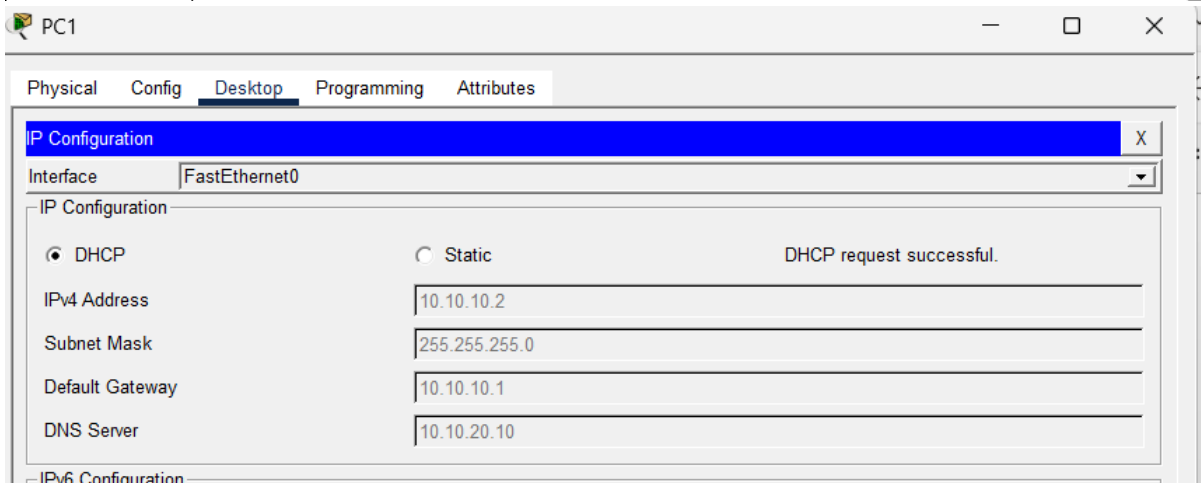
Router#show dhcp lease
Temp IP addr: 10.10.30.2 for peer on Interface: FastEthernet0/0
Temp sub net mask: 255.255.255.0
  DHCP Lease server: 10.10.30.1 , state: Bound
  DHCP Transaction id: 04A04CCC
  Lease: 86400 secs, Renewal: 43200 secs, Rebind: 75600 secs
Temp default-gateway addr: 10.10.30.1
  Next timer fires after: 11:58:25
  Retry count: 0 Client-ID:cisco-00D0.BCDD.2A01-Fa0/0
  Client-ID hex dump: 636973636F2D303044302E424344442E
                       24130312D4661302F30
  Hostname: Router
```

98.

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ip dhcp pool CAMPUS-LAN
^
% Invalid input detected at '^' marker.

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool CAMPUS-LAN
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)#exit
Router(config)#
```



```
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration   Type
                Hardware address
10.10.10.2      000B.BE89.59A4   --                 Automatic
Router#show ip dhcp pool

Pool CAMPUS-LAN :
Utilization mark (high/low)    : 100 / 0
Subnet size (first/next)       : 0 / 0
Total addresses                 : 254
Leased addresses               : 1
Excluded addresses             : 0
Pending event                  : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.10.10.1        10.10.10.1 - 10.10.10.254   1 / 0 / 254
Router#
```

```
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

99. removing R1's DHCP pool

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip dhcp pool CAMPUS-LAN
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip dhcp pool
Router#
```

Configuring the external server

DNS and DHCP Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: CAMPUS

Default Gateway: 10.10.10.1

DNS Server: 10.10.20.10

Start IP Address: 10.10.10.2

Subnet Mask: 255.255.255.0

Maximum Number of Users: 50

TFTP Server: 0.0.0.0

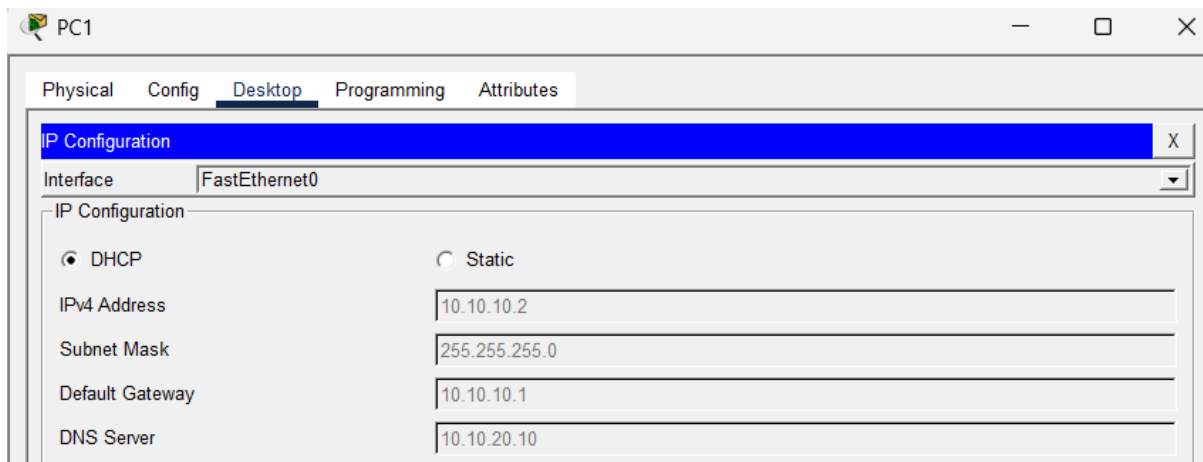
WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
CAMPUS	10.10.10.1	10.10.20.10	10.10.10.2	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.10.20.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Adding the IP helper address

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip helper-address 10.10.20.10
Router(config-if)#exit
Router(config)#
```

Setting PC1 to DHCP



Verifying on a PC

```
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.10: bytes=32 time=48ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 48ms, Average = 16ms

C:\>
```

On a router

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.10.30.2      YES DHCP    up          up
FastEthernet0/1 10.10.10.1      YES manual  up          up
FastEthernet1/0 10.10.20.1      YES manual  up          up
FastEthernet1/1 unassigned      YES unset   administratively down down
Vlan1           unassigned      YES unset   administratively down down
Router#show running-config
Building configuration...

Current configuration : 812 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
```

100.

An ip helper-address is needed when the DHCP server is on a different subnet because DHCP client requests are broadcast messages and routers normally do not forward broadcasts between networks. The ip helper-address tells the router to forward the DHCP request to the DHCP server on the other subnet.